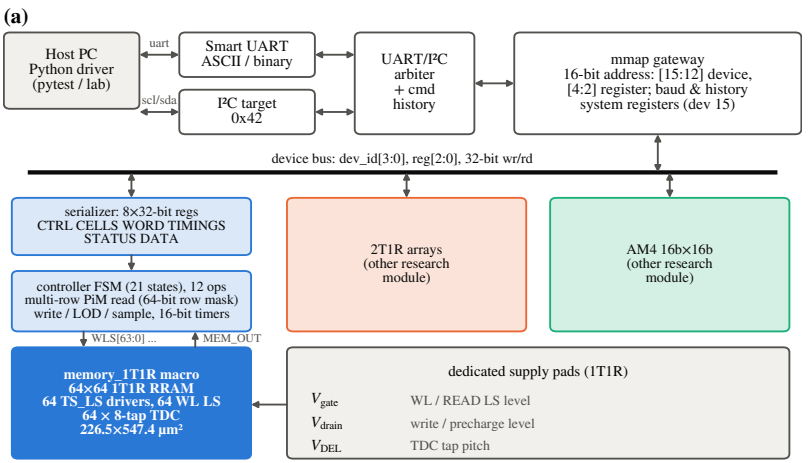
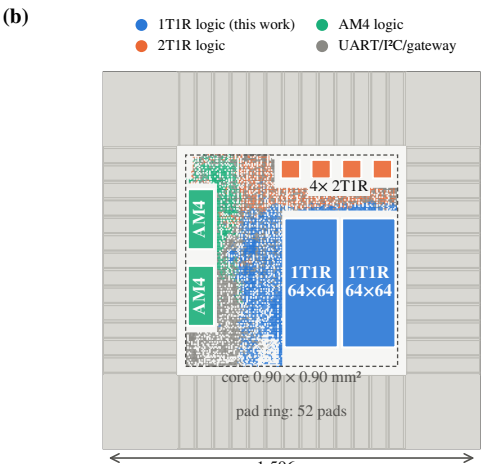

(a)
Host PC
Python driver
(pytest / lab)
uart
scl/sda
Smart UART
ASCII / binary
I²C target
0x42
UART/I²C
arbiter
+ cmd
history
mmap gateway
16-bit address: [15:12] device,
[4:2] register; baud & history
system registers (dev 15)
device bus: dev_id[3:0], reg[2:0], 32-bit wr/rd
serializer: 8×32-bit regs
CTRL CELLS WORD TIMINGS
STATUS DATA
controller FSM (21 states), 12 ops
multi-row PiM read (64-bit row mask)
write / LOD / sample, 16-bit timers
WLS[63:0] ...
MEM_OUT
memory_1T1R macro
64×64 1T1R RRAM
64 TS_LS drivers, 64 WL LS
64 × 8-tap TDC
226.5×547.4 µm²
2T1R arrays
(other research
module)
AM4 16b×16b
(other research
module)
dedicated supply pads (1T1R)
V_{gate} WL / READ LS level
V_{drain} write / precharge level
V_{DEL} TDC tap pitch


(b)
1T1R logic (this work)
2T1R logic
AM4 logic
UART/I²C/gateway
4× 2T1R
AM4
AM4
1T1R
64×64
1T1R
64×64
core 0.90 × 0.90 mm²
pad ring: 52 pads
1.596 mm